

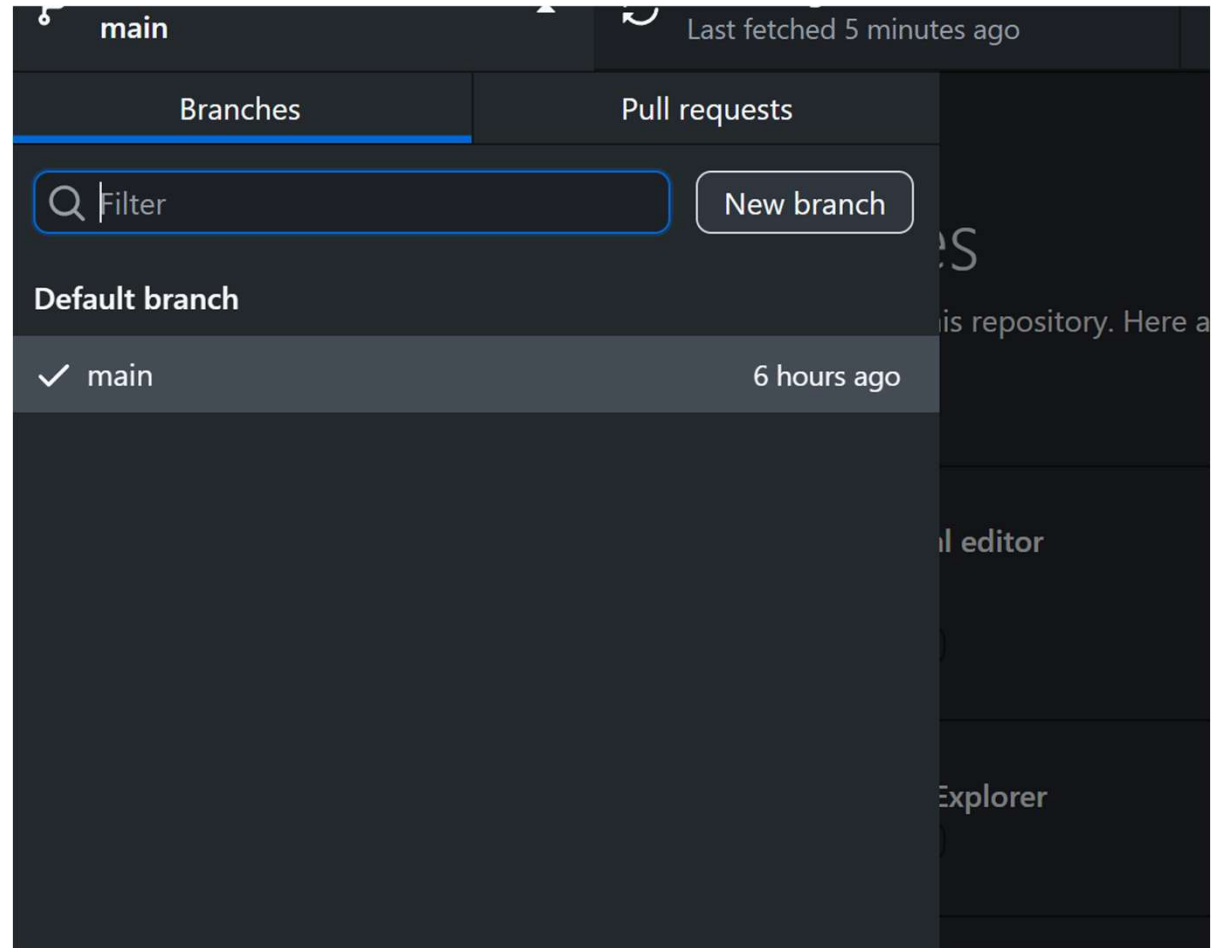
Graham's Guide to KiCad

Have fun (maybe)



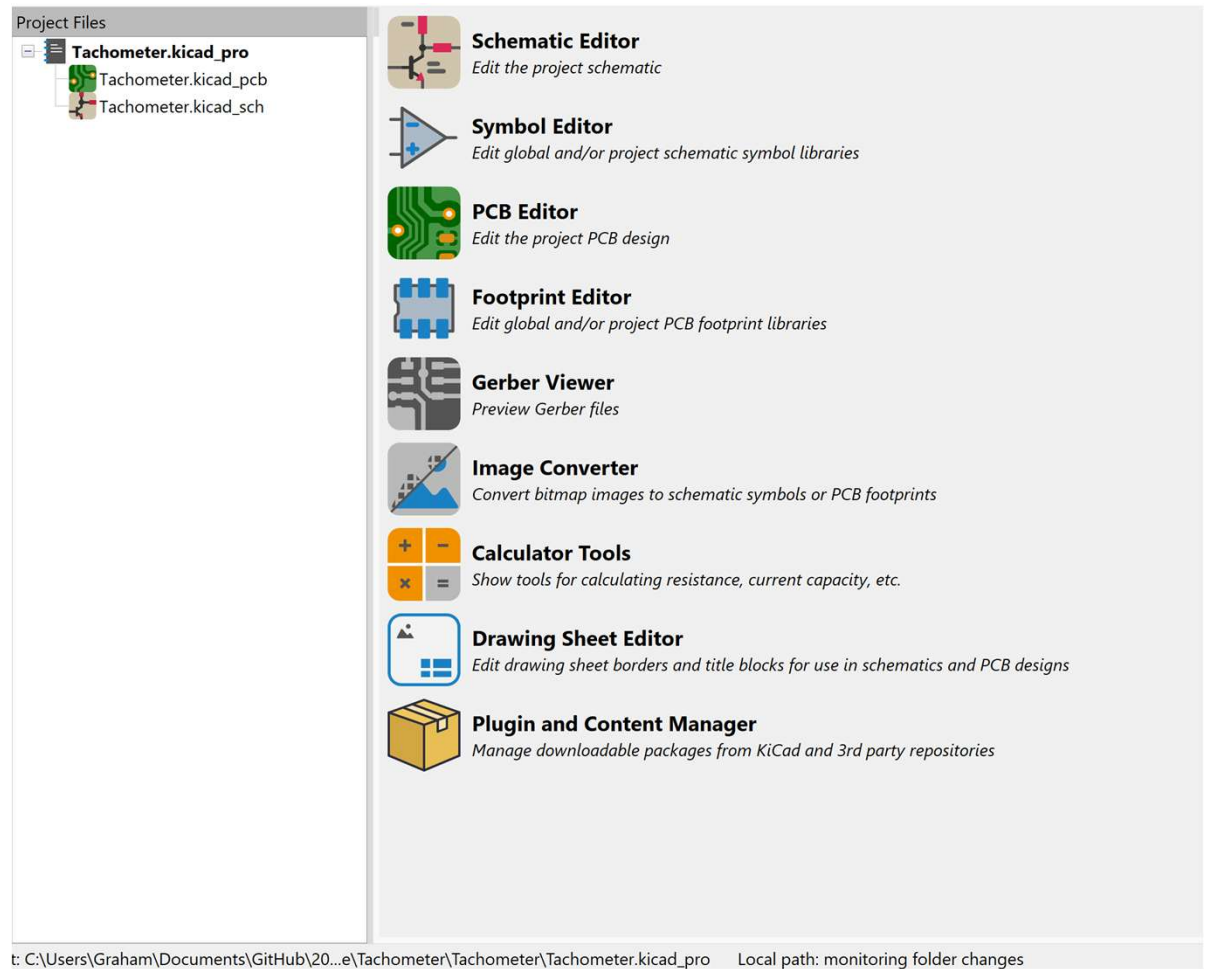
Branches

- Make sure you are on the Github branch dedicated to your sensor
- If no such branch exists just make one
- Name it after the sensor you are working on
- Please no nondescript names (Sensor 1 V1)



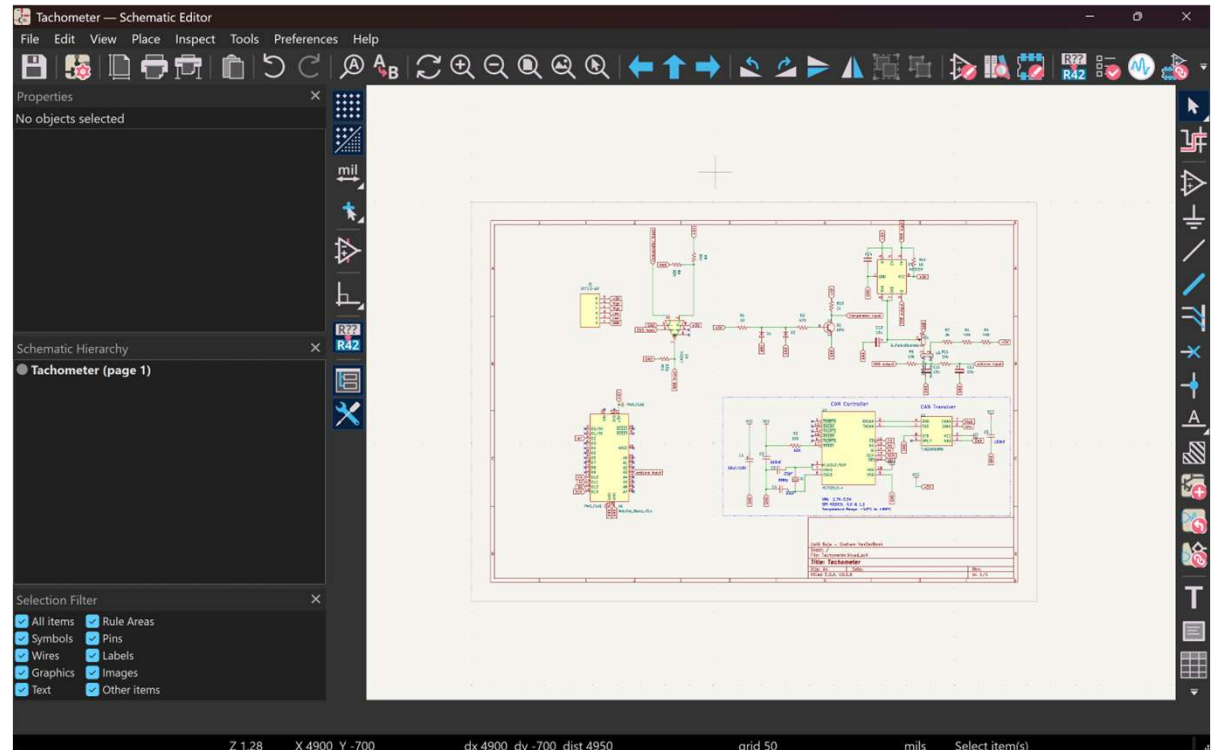
Creating your Project

- File (Top left) > New Project (First option)
- Make sure to save it to the 2026-2027 Hardware folder that corresponds to your project, if there is no folder for it make one in the Projects folder
- If you do not know where the projects folder is, it is probably Documents>Github>2026-2027 Hardware>Projects
- Name it sensibly (Ideally 26-27 [Insert Sensor name here])



Getting Started

- Click “Schematic Editor”
- It should look something like this
- Click things and play around
- This is where you create the schematic of your circuit and connect all your components before designing the PCB
- Many common problems on PCBs originate on the schematic

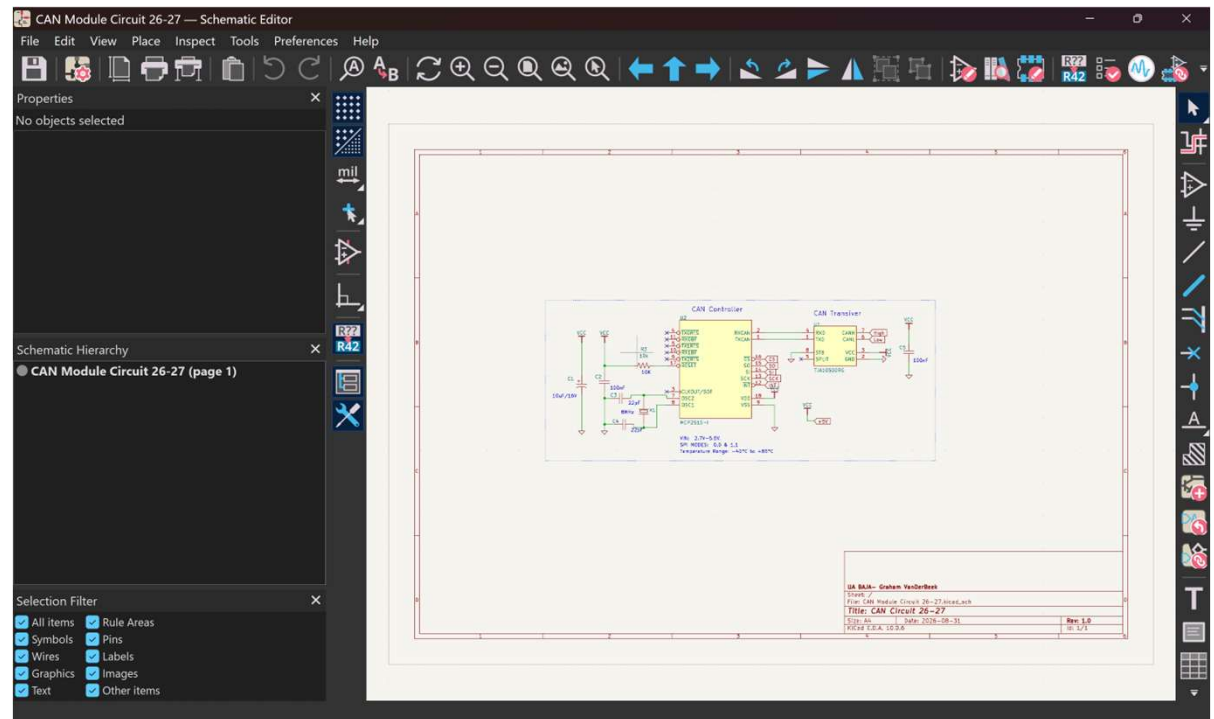


Features of Schematic Editor

- New KiCad 10 weirdness (Long Clicks)
- Wires, Global Labels (Tags), and Symbols can all connect components, but some are better in some circumstances.
- Place Symbols is how you add components
- Assign footprints (Make sure to save and apply before leaving)
- Place no connects (x) on any unused pins
- Text is nice to label different sections on the schematic

Step 1: CAN you do it?

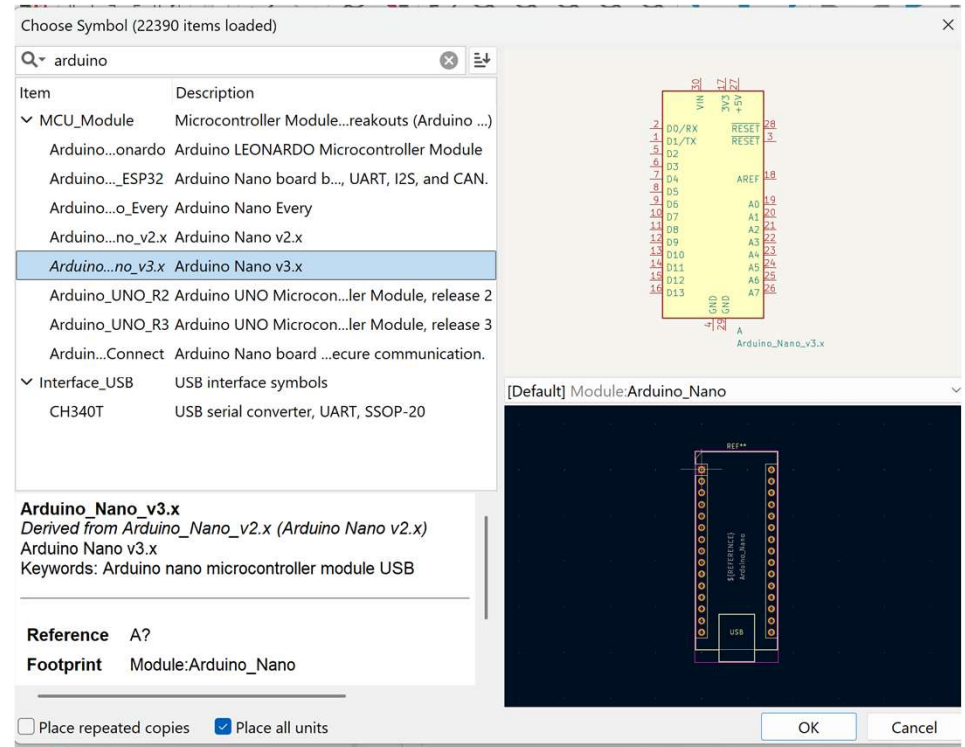
- With your sensor open
- Go to 2026-2027 Hardware>Shared Elements>CAN Module Circuit 26-27
- Then open the Kicad Project File and enter schematic editor
- Drag over the entire schematic and copy
- Return to your sensor, click on the central workspace, and paste the whole circuit
- This is one of the three main parts of most sensors



Step 2:

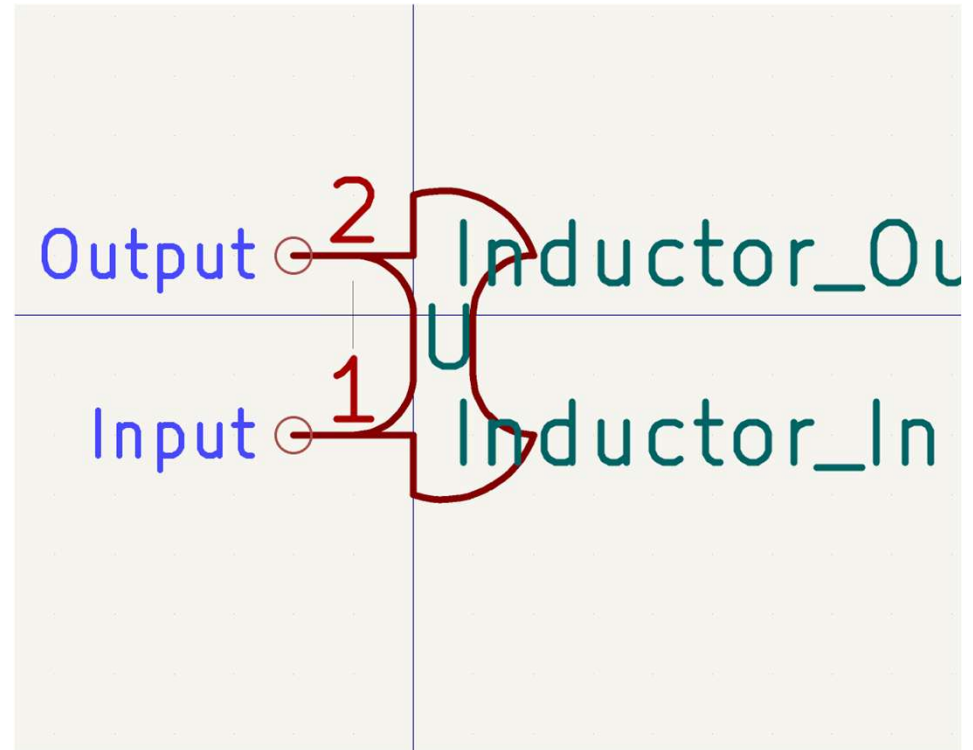
You do Arduino

- Use the symbol placing tool to place an Arduino nano (not Nano Every)
- This is the second main part of most sensors (Some may use an ESP32 but it's the same procedure)
- “Arduino_Nano_v3.x”



Step 3: Symbols

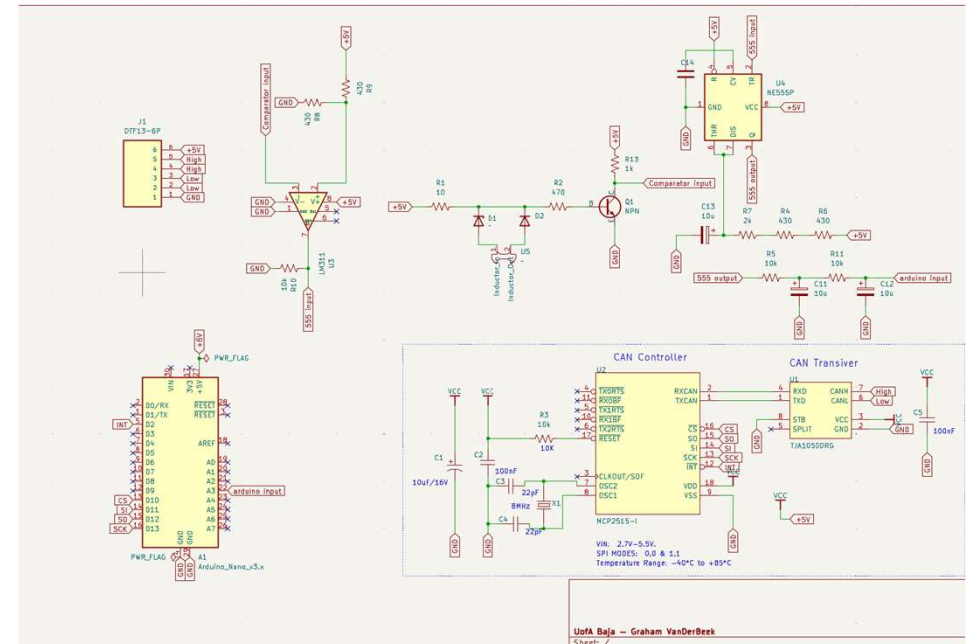
- Some sensors do not need a custom symbol and have a symbol already included if that is the case for your sensor skip ahead, otherwise this is how to create a custom symbol
- Click “Symbol Editor” on the top bar
- File>New Library>Project>Name it something descriptive>Save
- Find the new library in the left-hand bar select it and click new symbol>name it>OK
- Symbols need one pin per input or output then give it a fun shape (This will vary, ask or look at your sensor)



Step 4: Figure it out

Make the rest of your schematic, ensure it includes everything listed here. The example is more complicated than most so do not worry if yours is not this crazy.

- Make sure it has a DTF13-6P (Instructions are in the .zip file in shared elements) and use the pinout from the references file
- Make sure your custom symbol is there, and you have enough if your PCB has more than 1 sensor attached
- Make sure the Arduino is connected to the INT, CS, SO, SI, and SCK of the CAN controller (Pinout can be found in references)
- Make every pin is connected or has an X



Step 5: Footprints?

- Click “Assign Footprints” on the top bar and wait for it to load (It really is a race)
- Most should already be filled (The CAN module and hopefully the Arduino)
- If one is missing, you need to assign it (Like U5)
- If the DTF13-6P is missing, follow the instructions for adding the footprint in its folder
- If you do not know what a missing one should be, ask
- If you had to create a custom symbol, it also needs a footprint, instructions for that are on the next slide
- Make sure to click “Apply, Save Schematic, and Continue” if not you will need to restart this step

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1      A1 - Arduino_Nano_v3.x : Module:Arduino_Nano
2      C1 -      10uF/16V : Capacitor_THT:CP_Radial_Tantal_D5.0mm_P2.50mm
3      C2 -      100nF : Capacitor_THT:CP_Radial_D5.0mm_P2.50mm
4      C3 -      22pF : Capacitor_THT:CP_Radial_D5.0mm_P2.50mm
5      C4 -      22pF : Capacitor_THT:CP_Radial_D5.0mm_P2.50mm
6      C5 -      100nF : Capacitor_THT:CP_Radial_D5.0mm_P2.50mm
7      C11 -      10u : Capacitor_THT:CP_Radial_D5.0mm_P2.50mm
8      C12 -      10u : Capacitor_THT:CP_Radial_D5.0mm_P2.50mm
9      C13 -      10u : Capacitor_THT:CP_Radial_D5.0mm_P2.50mm
10     C14 -      C : Capacitor_THT:C_Disc_D6.0mm_W4.4mm_P5.00mm
11     D1 -      ~ : Diode_THT:D_A-405_P7.62mm_Horizontal
12     D2 -      D : Diode_THT:D_A-405_P7.62mm_Horizontal
13     J1 -      DTF13-6P : DTF13-6P:TE_DTF13-6P
14     Q1 -      NPN : Package_TO_SOT_THT:TO-92L_Inline
15     R1 -      10 : Resistor_THT:R_Axial_DIN0204_L3.6mm_D1.6mm
16     R2 -      470 : Resistor_THT:R_Axial_DIN0204_L3.6mm_D1.6mm
17     R3 -      10k : Resistor_THT:R_Axial_DIN0204_L3.6mm_D1.6mm
18     R4 -      430 : Resistor_THT:R_Axial_DIN0204_L3.6mm_D1.6mm
19     R5 -      10k : Resistor_THT:R_Axial_DIN0204_L3.6mm_D1.6mm
20     R6 -      430 : Resistor_THT:R_Axial_DIN0204_L3.6mm_D1.6mm
21     R7 -      2k : Resistor_THT:R_Axial_DIN0204_L3.6mm_D1.6mm
22     R8 -      430 : Resistor_THT:R_Axial_DIN0204_L3.6mm_D1.6mm
23     R9 -      430 : Resistor_THT:R_Axial_DIN0204_L3.6mm_D1.6mm
24     R10 -      10k : Resistor_THT:R_Axial_DIN0204_L3.6mm_D1.6mm
25     R11 -      10k : Resistor_THT:R_Axial_DIN0204_L3.6mm_D1.6mm
26     R13 -      1k : Resistor_THT:R_Axial_DIN0204_L3.6mm_D1.6mm
27     U1 -      TJA1050DRG : Package_SO:SOIC-8_5.3x5.3mm_P1.27mm
28     U2 -      MCP2515-I : Package_SO:SOIC-18W_7.5x11.6mm_P1.27mm
29     U3 -      LM311 : Package_DIP:DIP-8_W7.62mm
30     U4 -      NE555P : Package_DIP:DIP-8_W7.62mm
31     U5 -      ~ :
32     X1 -      8MHz : Crvstal:Crvstal SMD_HC49-SD

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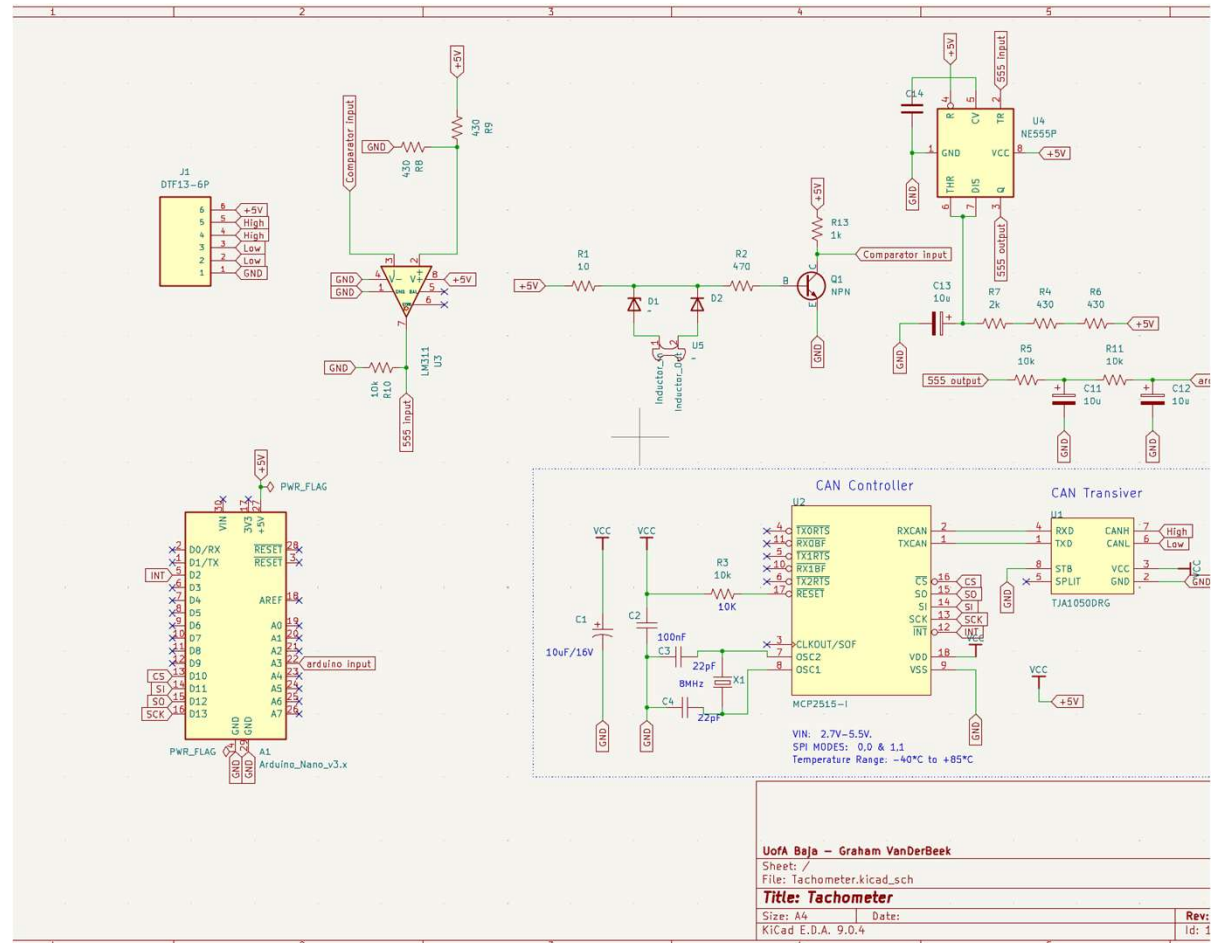
Custom Footprints??

- This part is going to be more specific to each sensor so it will really depend on exactly what each of your sensors requires
- From the schematic editor, click on “Footprint Editor” (Top bar)
- The easiest way is to use the footprint wizard (it has the red star on it)>Circular Pad Array>Then edit the parameters to match your sensor (You may need to drag the left bar to be bigger by hovering over the border and dragging
- Make sure drill is 2mm and diameter is 3mm to allow for soldering a wire into it. Then you may need to make the outline bigger to get rid of an error. Further make sure that the pads are not touching. If they are, make the center diameter larger
- Saving just requires clicking “Export to Footprint Editor” and then following similar procedures to the symbol (create library then save into new library)
- Now you can assign your custom symbol to your new footprint



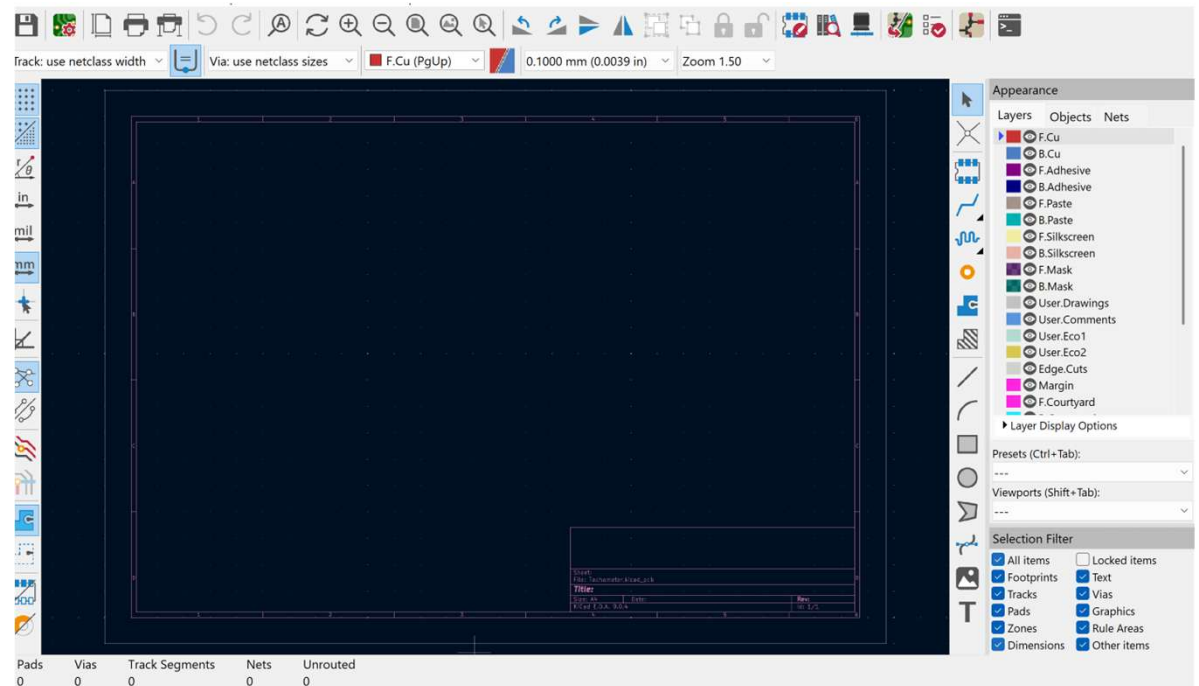
Congratulations!

- You have now made a complete schematic with footprints assigned to everything which means you are halfway done and ready to advance to making the PCB
- However, before you do that, show off, brag about your accomplishment, and have someone, ideally one of the leads, look at your schematic (It is a lot easier to fix mistakes on the schematic so a small thing caught here can save hours of work later)
- Once you have finished celebrating and a lead has checked out your schematic, if you dare to advance to the hard part of making the PCB (it's the fun part in my opinion) click the green "Switch to PCB Editor" button



The PCB Editor

- The PCB editor's biggest difference from the schematic editor is the bar on the right. This tells you what layer you are interacting with. These are the most important layers we will work with
 - F Cu – The front copper, red
 - B Cu – The back copper, blue
 - Silkscreen – Printed layer, top and bottom, cream/peach colored
 - Edge Cuts – Define the Edges of the board.
- Play with the features and see what everything does

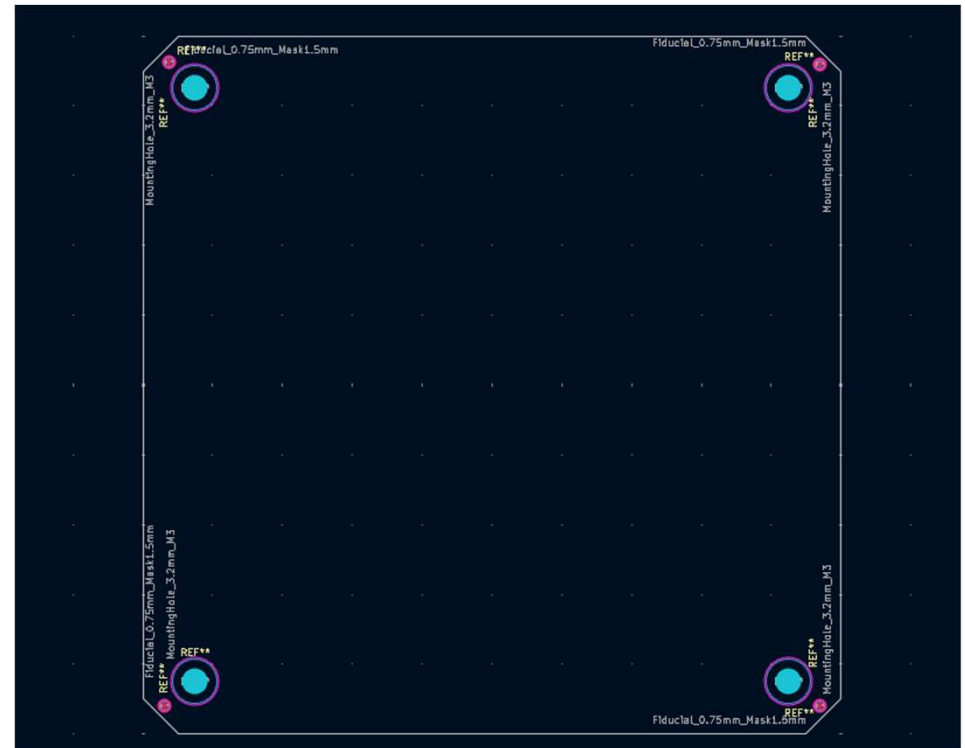


Features of the PCB Editor

- Footprints are components (They go with symbols from the schematic)
- Tracks connect footprints
- Vias connect from one side of the board to another
- You can control which layer of the board you are working on with the right-hand side bar
- Changing the resolution of the grid is helpful (Start big get small)
- Little blue lines tell you what connects to what (But they can be misleading if there is more than one component connected at the same time)
- Fill zones are perfect for GND but useful for other things (Regenerate them with b)

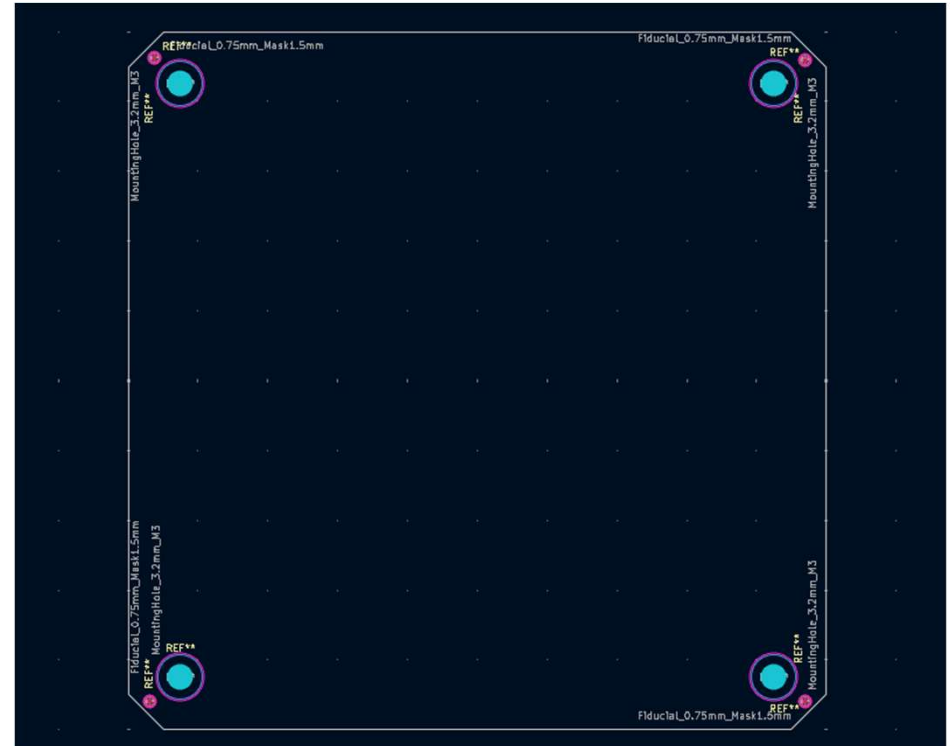
Step 6: Setting the board

- Start by selecting the edge cuts layer then the rectangle tool (also on right) and drawing a rectangle.
- Now right click on it, select properties and change its size to 50mm by 50mm (This is for convenience of designing enclosures later, some PCBs may be bigger)
- Next right click it again select shape modification and either chamfer or fillet the edges (pick whichever you prefer) at 5mm.
- Now to add mounting holes select place footprints and search “MountingHole_3.2mm_M3” place one of these in each corner with the center exactly 7.5mm from each size of the original rectangle. (A tip to do this it to use the dropdown at top center and increase the size to 2.5mm which effectively locks your movement onto a grid of 2.5mm by 2.5mm.)
- Yours will look a bit different from this example, as the example is for a larger PCB than the standard 50x50



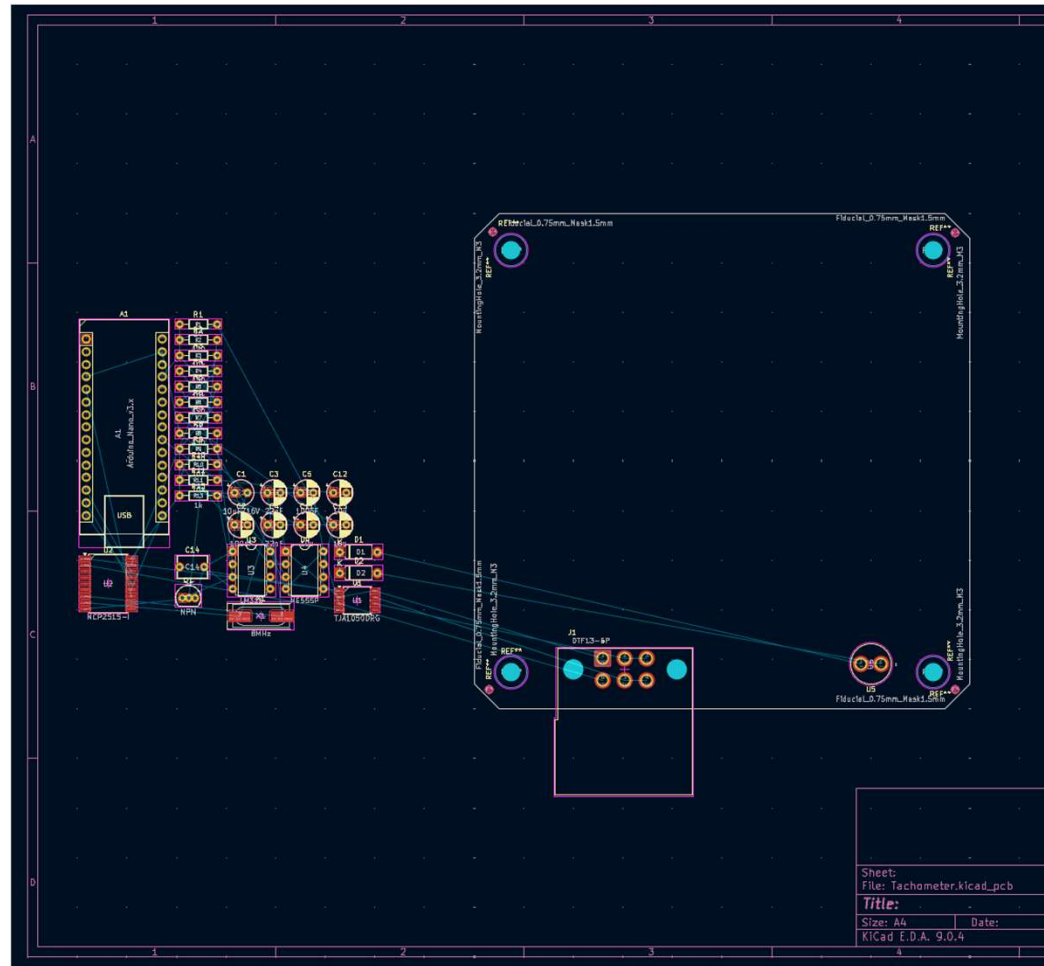
Step 6 cont.

- Repeating the procedure for the mounting holes instead search “Fiducial_0.75mm_Mask1.5mm” then place one of these in each corner between the corner of the board and the mounting hole. Exact position does not matter as much so just make sure they do not overlap with the mounting hole or go over the edge.
- In the end it should look something like this



Step 7: Layout

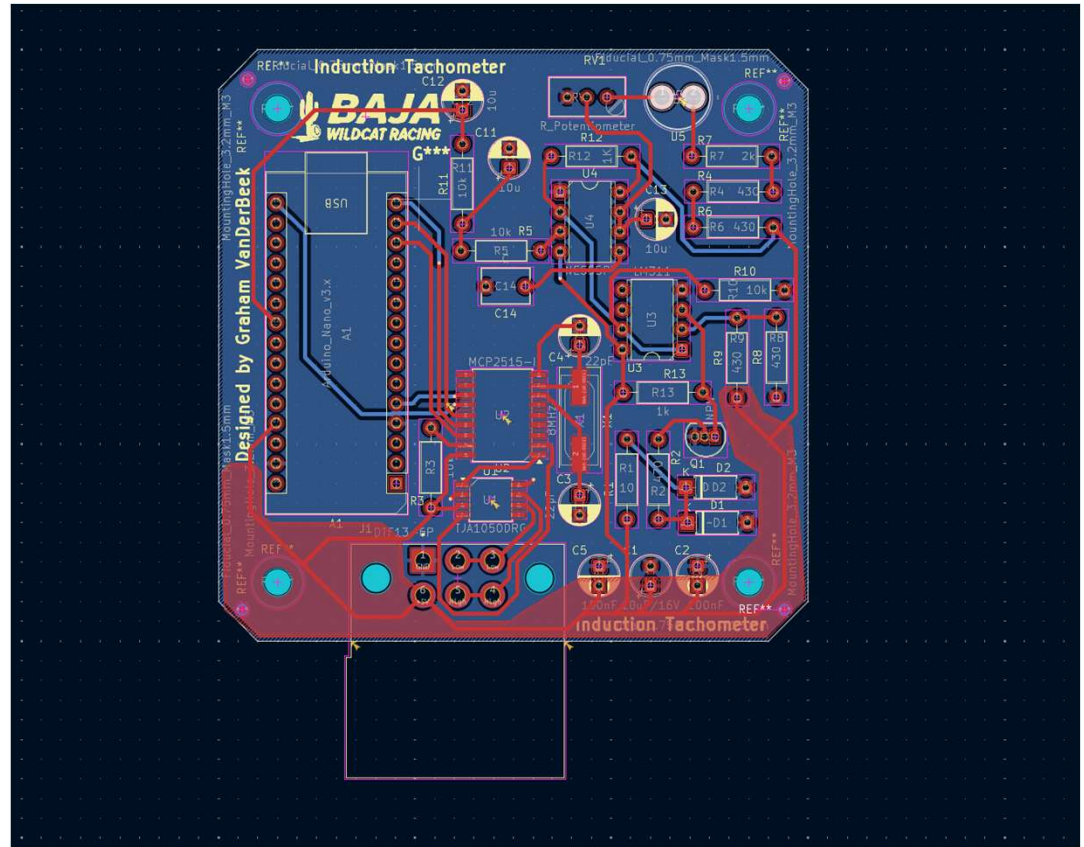
- The next step is to place the footprints onto the PCB, most of these can be placed wherever you like on the PCB, however a few have important locations.
- Click “Update PCB From Schematic” (top bar)> “Update PCB”> Then place the footprints by clicking on your workspace (Then I recommend placing your footprints off to the side of you PCB for clarity).
- Place the DTF13-6P so that the left mounting hole (the teal hole) is 20mm from the left edge and 8mm from the bottom edge. This can be a bit tricky to get exact, but it makes future work easier
- Then, if you needed to make one, your sensor’s custom connector should go on an edge.



Step 8: Figure it out Pt. 2

A few things to keep in mind:

- Make sure that the USB end of the Arduino is not blocked by other components.
- Try to keep related components grouped (CAN, Arduino, Sensor inputs)
- In general, thicker traces are better, but +5V traces must be at least 0.5 mm thick.
- Ground is best to be kept on the b.copper layer through a ground fill zone (ask or look it up)
- Vias are useful to get from one side of the board to another, but keep them to a minimum
- Try not to cross data traces with +5V traces
- Make sure to label your sensor, include your name, and the Baja logo (In shared elements)
- A +5V filler layer on f.copper nice but not required



PCB Complete!!

- Assuming you have completed all the previous steps, your PCB should be complete. Now have someone else check it out to see if you have missed anything. Then get one of the leads to get you ready for the official PCB review.
- As a last treat click the “3D Viewer” button on the top bar and check out your (probably incomplete) PCB 3D rendering.

